

Dublin City University - School of Computing

**BSc in Enterprise Computing
4th Year Project Proposal (CSC1118)
Idea Proposal
2024/2025**

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Project Title:

PulsePath: Festival Crowd Management

Date:

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1. Project Summary

Our mobile application, PulsePath, aims to change the way festivals operate and improve attendee experience by facilitating effective crowd management and communication. The application will provide real-time location and density data to end-users and display both available and overcrowded zones throughout large scale events, with a direct focus on festivals. By utilising existing site maps of festivals, [2] the application will adopt geolocation technology to track festival goers' current locations to represent crowd density levels at facilities within festival grounds on an interactive map. Amenities such as restrooms, food vendors and water filling stations will be shown on the map, combined with a user-friendly interface to represent low, medium and high crowd density levels. Festival goers can use our application to manage their time efficiently during the event, and also avoid overcrowded amenities which are tracked for them in real-time. In addition to real-time crowd density tracking, our mobile application will offer customisable alerts and notifications to enhance the user experience. Festival goers will have the option to set personalised preferences for receiving notifications about specific areas of the event, such as when a food stall, restroom, or stage becomes too crowded. These alerts will allow users to stay informed without constantly checking the app, helping them avoid congested zones and better manage their time during the festival. By providing these notifications, the app helps attendees to make smarter decisions and enjoy a more seamless, stress-free experience.

As project partners, we chose this idea because of our first-hand experiences with overcrowded and unsafe conditions at festivals in Ireland. We observed that existing crowd management technologies, such as Density.IO, focus primarily on large commercial spaces like airports or shopping centres, with limited attention to festivals. We saw a gap where festivals - particularly from the attendee and organiser perspectives - could benefit significantly from better crowd management. Our goal is to address inefficiencies in festival operations, including long waiting times for amenities like restrooms and food stalls, which can take away from the overall experience. Additionally, festival organisers often struggle to allocate resources effectively, leaving some areas under-monitored and potentially unsafe for attendees [3]. By developing a real-time crowd tracking application tailored to festivals, we aim to improve both the attendee experience and the organisers' ability to ensure safety and efficiency throughout large-scale events.

We believe that a data-driven approach to crowd control would be highly beneficial to improving event safety practices, as new festivals are emerging and growing their fan base. The potential for data insights is a key reason we chose this project. By collecting and analysing real-time crowd movement data, festivals could benefit from much more than just immediate crowd control solutions [4]. This data could provide valuable insights into how attendees move throughout the event and which areas are most and least popular. Organisers could then use this information during post-event analysis to identify areas for improvement, such as adjusting layouts, increasing staff presence in key locations, or better managing resources like food stalls and restrooms. Over time, this data-driven approach would enable festivals to plan future events more effectively, improving both safety and the overall attendee experience. Our idea proposes a long term solution that helps organisers make informed decisions based on real world data, going beyond real-time monitoring to contribute to strategic event planning.

2. Expected Technical Delivery

For the technical delivery of our application, we aim to provide a full suite of functionalities and ensure a seamless user experience. Our primary technical goal is to integrate geofencing technology [5], to create virtual boundaries around key areas of the festival (e.g., tents, stages, restrooms). When users with the app enter or leave these areas, the system will be able to track their movements in real time and update the count for the capacity within each geofence. The app will leverage the user's location data (with permission) to monitor the number of attendees in each geofenced area. We plan on collaborating with festival organisers to allow us to solely display the festival line-up through our app, thus gaining downloads from the majority of the attendees and collecting enough data to make an accurate monitoring of the festival's live crowd capacity.

One of our key deliverables will be our user-friendly interface, where attendees will be able to view the live capacity status of the festival's different zones on an interactive map of the festival grounds. We're planning on using React/React Native to develop our frontend in order to display the geo-fenced areas with colour-coded markers indicating the current crowd density in each zone (e.g., green for low, red for high). The app will feature a navigable map, helping users find less crowded areas in real-time.

We're planning on developing our backend using either Node.js or Django and providing the API endpoints for managing user location data, fetching the real-time capacity updates, and handling geofence triggers. The backend will be designed to handle multiple users simultaneously and securely store our data. As for our database, we plan to make use of a relational database like PostgreSQL or MySQL to store all of the event data, user movements, geofence coordinates and historical crowd density. We plan on also logging the location and movement data for reporting and analysis purposes.

Adhering to the festival organisers' needs, we plan to develop an admin portal, accessible via web, where they can monitor the live status of geofenced areas. This might include visual heat maps and real-time reporting on capacity across multiple zones, in order to give the organisers the ability to act preemptively in case of overcrowding that could lead to hazardous incidents. In addition to real-time monitoring, we also plan on providing post-event analytics on this organiser dashboard, to help them evaluate crowd movements, peak times and overall event flow, allowing them to optimise future festivals.

Security and compliance is one of our main priorities and we plan on implementing user authentication using services like Firebase Authentication or Auth0, to ensure data security and privacy for all our app users. Our app will strictly adhere to GDPR regulations with end-to-end encryption.

Prior to submitting, we would like to conduct some rigorous testing to make sure our system works as expected. We might create mock geofences using existing coordinates and mock users that will enter and exit them at different times and check how our system responds to them. We plan on developing and testing our app to be compatible with both iOS and Android devices.

While our initial delivery will focus on one event/festival, we want to develop this app with scalability in mind, allowing for easy replication across multiple festivals and venues in the future.

By the end of this project, we expect to deliver a user-friendly web and mobile application that allows festival-goers to track real-time capacity at various locations within an event. This app will not only

enhance the user experience by reducing overcrowding but will also give event organisers valuable insights into crowd management, enabling safer and more efficiently run events.

3. Market Rationale

Our festival crowd management application has a wide and expanding market due to the global rise in popularity of large-scale music, arts, and cultural festivals. Massive crowds attend these events, and festival organisers are always looking for creative ways to keep attendees happy, safe, and comfortable. [1] Recent studies indicate that the festival sector is expected to develop significantly in the near future, with Europe being a leading destination for outdoor events. Better crowd control tools that can assist organisers and attendees in navigating these challenging circumstances are becoming more and more necessary as a result of the increase in attendance. Our idea aims to satisfy this increasing demand by offering user-friendly navigation, geolocation features, and real-time data on crowd density.

Festival attendees and festival organisers are our application's two main user segments. The app is appealing to attendees who wish to make the most out of their festival experience by avoiding busy areas and cutting down on the amount of time they have to wait for necessities like food stalls and restrooms. Our software enables users to better manage their time and enjoy the festival without stress or frustration by giving real-time information into crowd numbers at common areas within the festival grounds. The long duration of festivals and their large outdoor spaces require easy navigation and real-time updates to enhance the attendee experience.

The application provides useful features for organisers to increase operational efficiency. The management of resources like food, water, and restrooms, as well as the assurance of sufficient safety and security, provide many difficulties for festival organisers. By giving event planners access to real-time crowd density data, our app helps them better manage personnel and resources, divert people to less used spaces, and decide on security and emergency response protocols. Through the integration of our app, event organisers may enhance crowd management, minimise traffic, and establish a more secure atmosphere for participants. Post-event data analytics will give event planners knowledge about the behaviour and flow of the crowd, enabling them to make well-informed adjustments for future events.

Our festival crowd management application will use a combination of primary and secondary research approaches to validate the market. In order to determine the problems festival goers and organisers have with crowding, wait times, and resource management, our main research will involve surveys and interviews with these parties. This will enable us to better adapt the app's features, like geolocation services and real-time crowd notifications, to the needs of the user base. As the organisers, we plan to receive input in real-world festival situations by conducting interviews and testing the app during small-scale pilots. We will examine competitors' applications and market data for secondary research purposes in order to identify any gaps and trends in the event management industry.

These observations will guarantee that our software is positioned to satisfy the increasing need for improved crowd control methods, enhancing the experience of both festival goers and organisers.

4. Proposed Timeline

October 21 - November 1, 2024 (Dates to be confirmed)

- **Task:** Present project proposal to Panel
- **Responsibility:** Zeynep Altintas and Lizel Sulaj
- **Details:** Present the project idea to the panel and gather feedback for improvements. Ethical and Idea Proposal forms will be submitted prior to this.

November 2 - December 3, 2024

- **Task:** Development of mobile application, conduct small sample surveys
- **Responsibility:** Zeynep Altintas and Lizel Sulaj
- **Details:** Start working on the application's core features, focusing on real-time crowd management functionalities. Begin with surveys with smaller samples to prepare market research.

December 4 - December 17, 2024

- **Task:** Conduct User Surveys and Interviews
- **Responsibility:** Zeynep Altintas and Lizel Sulaj
- **Details:** Gather insights from potential users, including festival attendees and organisers, to refine the app's features.

December 18 - January 17, 2025

- **Task:** Mid-Term Delivery Documentation Preparation and Submission
- **Responsibility:** Zeynep Altintas and Lizel Sulaj
- **Details:** Compile all mid-term documentation and upload it to the group repository.

January 18 - March 15, 2025

- **Task:** Continue Development and Testing
- **Responsibility:** Zeynep Altintas and Lizel Sulaj
- **Details:** Refine app features based on feedback, focusing on improving user experience and performance.

March 15 - April 18, 2025

- **Task:** Preparation of Final Deliverables, Submission
- **Responsibility:** Zeynep Altintas and Lizel Sulaj
- **Details:** Complete all project deliverables, including the final version of the app and necessary documentation. Ensure that all project materials, including source code and PDFs, are uploaded to the repository.

May 5 - May 16, 2025

- **Task:** Group Interview/Project Presentation and Project Expo
- **Responsibility:** Zeynep Altintas and Lizel Sulaj

- **Details:** Prepare for and attend the CSC1118 demos and project expo, presenting the application and findings.

5. Workload Distribution

Lizel Sulaj:

Backend Development:

- Design and implement the backend APIs using [Node.js/Django](#). This includes building endpoints to enable efficient user management, location updates, and geofencing alerts.
- To ensure that crowd and capacity data is updated instantly and real-time alerts and notifications will be sent out across multiple users efficiently, I plan on implementing real-time data handling through the use of [Socket.io/WebSocket](#).

Databases Development:

- Set up a secure and scalable database with PostgreSQL/My SQL to store user data, geofence details, and other event specific metrics.

Security and Authentication:

- Implement a secure user authentication using Firebase Authentication/Auth0 .

Geofencing and Location Services

- Define the geofences around event areas using services like Google Maps API and develop logic that will trigger updates when users location enters or exits these geofenced zones.
- Set up the geofencing trigger that will monitor the number of users within a specific geofence at a time.

Data Analysis

- Collect and analyse data collected from surveys/interviews
- Refine ideas regarding app functionality and features based on user feedback.

Zeynep Altintas:

Data Analysis:

- Analyse data collected from primary and secondary research, including interviews and surveys.
- Refine ideas regarding app functionality and features based on user feedback.

Business Strategy:

- Identify and propose enhancements to improve the overall user experience.
- Develop a comprehensive business plan that aligns with project goals.

Cloud Computing:

- Research and determine the most suitable cloud provider for the application (e.g. AWS tools)
- Implement cloud solutions that ensure high availability and scalability for the app.

Front-End Development:

- Design and implement a user-friendly interface that enhances the overall user experience of the application.
- Ensure that the app is intuitive and easy to navigate, allowing users to access key features effortlessly.

Collaborative Efforts

Regular Meetings:

- Schedule bi-weekly meetings to discuss progress, challenges, and updates on each component.

Testing and Optimisation

- Conduct stress testing on the backend to ensure it can handle high loads.
- Optimise for response time
- Ensure compatibility with both web and mobile platforms as well as Android and iOS
- Resemble geofencing functionality with mock coordinates to test our logic

Final Integration:

- Work together on the final app integration, ensuring seamless functionality between business insights and technical execution.

6. Staff Consulted

Our project supervisor is Paul M. Clarke.

7. References

- [1] Phagare, M. (2024). *Europe music festival industry report 2024*. Cognitive Market Research. <https://www.cognitivemarketresearch.com>
- [2] Patel, D. (2024). *Music Festival Market: Global Industry Analysis*. Growth Market Reports. <https://growthmarketreports.com>
- [3] Crowd planning for public safety. (2017). *Perspectives in Public Health*, 137(1), 25-28. <https://doi.org/10.1177/1757913916681265>
- [4] InEvent. (2023). *8 reasons why your event management should be data-driven*. <https://inevent.com>

- [5] Android Developers. (n.d.). *Geofencing overview*.
<https://developer.android.com/develop/sensors-and-location/location/geofencing>